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标题	Ru(II)-Catalyzed Asymmetric Transfer Hydrogenation of α-Alkyl-β-Ketoaldehydes via Dynamic Kinetic Resolution
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摘要（原文节选）

The (R,R)-Teth-TsDPEN-Ru(II) complex promoted the one-pot double C=O reduction of α-alkyl-β-ketoaldehydes through asymmetric transfer hydrogenation/dynamic kinetic resolution (ATH-DKR) under mild conditions. In this process, ten anti-2-benzyl-1-phenylpropane-1,3-diols (85:15 to 92:8 dr) were obtained in good yields (41 – 87%) and excellent enantioselectivities (>99% ee for all compounds). Notably, the preferential reduction of the aldehyde moiety led to the in situ formation of 2-benzyl-3-hydroxy-1-phenylpropan-1- ⋯

数据表 · Table 1（10 行，含数值行 10 行）

Synthesis of the α-benzyl-β-ketoaldehydes (rac)-10 from chalcones (8).

Entry	Structure	8 a	R1	R2	R3	13 b	Yield (%) c	10 d
1		8a	H	H	H	13a	87	10a
2	8b	H	H	OMe	13b	53	10b	48:52
3	8c	H	H	CF3	13c	67	10c	0:100
4	8d	H	H	F	13d	65	10d	15:85
5	8e	H	F	H	13e	79	10e	12:88
6	8f	F	H	H	13f	37	10f	24:76
7		8g	OMe	-	-	13g	92	10g
8	8h	Br	-	-	13h	91	10h	37:63
9		8i	Cl	H	F	13i	77	10i
10	8j	OCH2O	F	13j	88	10j	50:50	91

脚注（逐字）：a Obtained in a 6 mmol scale from the aldol condensation between the corresponding commercial acetophenones and aldehydes using KOH (7.5 equiv) in MeOH/H2O (1:1), rt, 2 – 27 h (65 – 96% yield). b Obtained in a 3 – 6 mmol scale. c Isolated yield after purification by column chromatography. d Obtained as a

正文对应句（逐字）：To enhance the conversion of compound 11j into anti-12j diastereoisomer, we raised the reaction temperature from room temperature (rt) to 45 ° C. This change led to a substantial conversion of the intermediate ketone 11j (11j/12j 7:93) (Table 2, entry 7). However, the higher yield of 12j at 45 ° C was

数据表 · Table 2（17 行，含数值行 17 行）

Optimization of the conditions for the ATH-DKR of (rac)-10j a.

Entry	[H] Source (Ratio) b	(R,R)-C-I (mol%)	Solvent c	Add. d	T (° C)	t (h)	11j/12j e	12janti/syn e
1	HCO2Na	D (2)	DCE/H2O	CTAB	rt	16	>99:1	nd
2	HCO2H/Et3N (5:4)	D (2)	DCE	-	rt	16	>99:1	nd
3	HCO2H/Et3N (5:4)	D (2)	MeCN	-	rt	16	68:32	84:16
4	HCO2H/Et3N (5:4)	D (2)	THF	-	rt	16	62:38	93:7
5	HCO2H/Et3N (5:4)	D (2)	EtOAc	-	rt	16	54:46	92:8
6	HCO2H/Et3N (5:4)	D (2)	PhMe	-	rt	16	44:56	92:8
7	HCO2H/Et3N (5:4)	D (2)	PhMe	-	45	16	7:93	87:13
8	HCO2H/Et3N (5:4)	C (2)	PhMe	-	45	16	85:15	59:41
9	HCO2H/Et3N (5:4)	E (2)	PhMe	-	45	16	95:5	nd
10	HCO2H/Et3N (5:4)	F (2)	PhMe	-	45	16	93:7	nd
11	HCO2H/Et3N (5:4)	G (2)	PhMe	-	45	16	>99:1	nd

Entry	[H] Source (Ratio) b	(R,R)-C-I (mol%)	Solvent c	Add. d	T (°C)	t (h)	11j/12j e	12janti/syn e
12	HCO2H/Et3N (5:4)	H (2)	PhMe	-	45	16	>99:1	nd
13	HCO2H/Et3N (5:4)	I (2)	PhMe	-	45	16	>99:1	nd
14	HCO2H/Et3N (5:4)	D (2)	PhMe	Cu(OTf)2	45	16	6:94	88:12
15	HCO2H/Et3N (5:4)	D (2)	PhMe	TfOH	45	16	4:96	85:15
16	HCO2H/Et3N (5:4)	D (4)	PhMe	-	rt	16	19:81	92:8
17	HCO2H/Et3N (5:4)	D (4)	PhMe	-	rt	24	20:80	90:10

脚注（逐字）：a All reactions were carried out on a 0.085 mmol scale using (R,R)-C-I, HCO2Na or HCO2H/Et3N as [H] source in the quantities indicated in each case. b HCO2Na (6 equiv.) and to F/T 7.6 and 6.1 equivalents of each were used, respectively. c At 0.25 M, using the pure solvent or in a 1:1 mixture with wa

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